

Group Project - Problem Statement

Our team will design a web based application which allows small independent artists to interact with their fan communities.

The system will act as a hybrid social media/art hosting site allowing the artists to upload their music, art images, and stories, message their fans, and post social media style videos.

Our team will investigate similar platforms such as soundcloud and instagram and decide what features we deem necessary to implement into our own project.

Our project will include:

- Multiple user roles: artist, user, and admin all with different permissions.
- Advanced business logic
- A database to handle persistence
- Workflows for artists, users, and admin to set up accounts as well as upload their media and interact with their fanbases.
- We plan to implement 3 tier architecture with a frontend webserver, backend application layer, and a database.
- We will implement a CI/CD pipeline as well as mandatory pull requests.
- We will deploy to a free web hosting service such as GCP, AWS, or Azure.

Core Roles

The following details the different user roles for our application.

Fan

A fan is a person who interacts with the artist's page, supports the artist, and attends their events/shows. Typical responsibilities may include:

- Maintaining an "about" style profile where they can show off the kinds of art they like and who they support.
- Explore new artists they may be interested in: they should be able to set a radius to see who is approximately near vs far from them.
- Following and interacting with Artists' posts.
- Messaging system for fans to message artists and each other.

Artist

An artist is a person who uploads a form of media for fans. Typical responsibilities may include:

- Maintaining an artist profile.
- Promoting their events and shows on their profile.
- Recording videos interacting with their fans.
- Messaging their fans or each other.
- Monitoring their comment sections, reporting inappropriate comments.

Administrator

An Administrator handles system functions that artists and fans should not have access to.

Typical responsibilities may include:

- Managing user accounts and access.
- Reviewing Artists applications to make sure they are not a fake artist.
- Viewing system diagnostics to ensure proper operation.
- Responding to artist reports and banning users if their actions go against the terms of service (ToS).

These descriptions are a general guide for us, we plan to be creative and go beyond our basic outline. Further we will create:

- Functional requirements
- Nonfunctional requirements
- Business rules
- Use cases
- Application workflows
- Data entities and relationships
- Authorization rules
- User interface behavior
- Appropriate exceptions and failure conditions

AI Related Functionality

Time dependent we hope to include bonus AI related functionality such as:

- AI music and image detectors to prohibit non-human made art from our platform:
 - <https://github.com/guyfloki/ai-image-detector>
 - <https://github.com/lofcz/ai-music-detector>
- Some form of audio and image graph database to power our recommendation engine based on art similar to art users currently interact with.
- AI based content moderation to limit hateful and inappropriate comments and content.